TEAM HANDBOOK · V1

AquaGrid

How every part of the system works, what each file is responsible for, and the things you must not “fix”.

DreamHacks 2026 · Track 2 — AI, Automation & Logic

github.com/sbongula/aquagrid

sbongula.github.io/aquagrid

Expo SDK 57 · React Native 0.86 · scikit-learn · Groq

00

What this document is

Read section 01 and 02. Read the rest when you touch that file.

AquaGrid decides, hour by hour, whether to run an island's desalination pump. It does that using a trained demand forecaster, a live weather forecast, a battery, the rain, and a set of rules that trade diesel against the risk of running the tank dry. It then explains every decision in plain language and watches for four kinds of failure.

The whole thing runs **on the phone with no server.** That is the single most important architectural fact and section 02 explains why.

00b

The problem, and how it is solved today

Read this before you explain the project to anyone.

On a small isolated island, water and energy are the same problem. Fresh water comes from rain when it falls and from a desalination plant when it does not, and that plant is usually the single largest electrical load on the island. The electricity comes from solar when the sun is up and from a diesel generator when it is not — and the diesel arrives by barge, every few weeks, whether or not you are running low.

So every hour there is a real decision: make water now, or wait. Making it at noon on a clear day is nearly free. Making it at 02:00 costs imported fuel. And the tank is a battery — it will hold a day or two of demand — which means the decision can be deferred, if you know what tomorrow looks like.

Almost nowhere is that decision actually made. It is delegated to a timer set decades ago.

Eight approaches in common use

None of these is stupid. All of them are blind in the same way.

Wall timer / fixed schedule The pump runs the same hours every day, typically overnight, inherited from mainland off-peak electricity tariffs.	On a diesel-and-solar island there is no off-peak. Those hours are the ones with no sun, so the timer systematically picks the most expensive water of the day.
Float switch / level control A level sensor starts the pump below a set point and stops it above another.	Purely reactive. It responds to the tank after it has already drained, never anticipates, and is equally happy to run at midnight as at noon.
Operator judgement with SCADA A skilled operator watches gauges and decides by experience when to run.	Works, and often works well — but it does not scale, does not survive staff turnover, and no human runs a 12-hour lookahead over a weather forecast every hour.
Commercial microgrid controllers Hybrid energy-management systems that dispatch generators, batteries and PV.	They optimise the electrical side, treating the desalination plant as a fixed load to be served. The water tank's value as storage is invisible to them. They also assume connectivity and a budget neither of which a village of 1,200 has.
Rainwater, managed separately Household and community tanks collect rain independently of the plant.	The plant makes water on days the sky is about to fill the tank for free, and the rain overflows while diesel burns.
Threshold alarms for leaks Alert when pressure or level crosses a fixed bound.	A threshold cannot distinguish a burst pipe from a busy evening. By the time the level is visibly wrong, the water is gone. Large utilities use district-metered minimum-night-flow analysis; that infrastructure does not exist on a small island.
Calendar or run-to-failure maintenance Membranes cleaned on a schedule, or when performance becomes obviously bad.	Too early wastes a clean; too late means months of quietly elevated energy per cubic metre, paid for in diesel.
Fuel tracked by dipstick Litres remaining checked manually against a barge schedule on a whiteboard.	Nobody projects the current burn rate forward against the delivery date until it is close, and the barge does not come early because you ran out.

What every one of them is missing

Each of these is reasonable on its own. The failure is that **none of them share a demand forecast**, so none can defer a decision. The timer cannot know that tomorrow is sunny; the float switch cannot know that rain is coming in four hours; the leak alarm cannot know what the tank *should* be doing at 03:00 on a Tuesday; the maintenance calendar cannot see that specific energy has been climbing for six weeks.

Predict demand once, and every one of those problems becomes tractable with arithmetic. That is the whole thesis of AquaGrid: **one forecast, shared by the scheduler and every detector.**

01

Run it in five minutes

Two halves: Python trains, the app consumes.

Train the model and export the forecast

```
cd ml
python -m venv ../.venv
../.venv/bin/pip install -r requirements.txt
../.venv/bin/python generate_data.py    # 17,520 rows of synthetic history
../.venv/bin/python train_model.py      # trains, evaluates, writes ../app/assets/forecast.json
```

`train_model.py` prints MAE, R^2 and feature importances, fetches a live 48-hour forecast from Open-Meteo, and writes the JSON the app ships with. If the network is blocked it falls back to a modelled clear-sky curve and says so in `weather_source`. **The build never blocks on the network.**

Run the app

```
cd app
npm install
cp src/config.example.js src/config.js   # optional: paste a Groq key
npx expo start                            # add --ios for the simulator
```

Install **Expo Go** on a phone and scan the QR code.

Verify the logic without a phone

```
cd app && node src/logic/test.js
```

Prints all three strategies, the savings, the leak result and the lookup-table fidelity. Exits non-zero if any check fails. Everything in `src/logic/` is free of React imports specifically so this works.

> IF THE PHONE CONNECTS BUT NEVER LOADS
>
> Campus and guest wifi block device-to-laptop traffic. Use `npx expo start --tunnel`.

> AFTER CHANGING A FUNCTION SIGNATURE IN LOGIC/
>
> Fast Refresh keeps stale module closures and you will get errors like `Cannot create property 'socKwh' on number` even though the code is correct. Restart with `npx expo start --clear`. This already bit us once.

> NEVER
>
> Do not run `npx expo prebuild` or `eas build`. Expo Go only. There is no time and no need.

02

Architecture, and why there is no backend

Read this before you propose adding an API.

Python trains and evaluates the model on a laptop and exports everything the app needs into one JSON file. The app bundles that file as a static asset. Scheduling, rainwater harvesting, battery dispatch, leak and asset detection, and all charting run in JavaScript on the device.

Why this and not a server

A laptop-hosted API reached from a phone means LAN addresses, ngrok tunnels, CORS and cold starts. Those are the four most common ways a demo dies. Removing the server removes all four.

More importantly it is the *correct* design for the problem. An island whose uplink is a satellite dish cannot depend on a cloud round-trip to decide whether to run its pump. **The app works in airplane mode.** Say that out loud in the demo.

> IF A JUDGE ASKS “IS THE AI REAL?”
>
> Yes. The model is trained and evaluated in Python on a chronological split, the metrics are in the README, and its predictions drive every decision. “Trained offline, inference on-device” is how most production mobile ML actually ships. Since we also export the model as a lookup table (section 10), the forecaster genuinely runs on the phone for locations the laptop never saw.

What is and is not synchronous

All scheduling is **synchronous** and lives in `useMemo`. Only the two LLM calls are async and they live in `useEffect`. Never mix them — putting a promise in `useMemo` is the classic way to break this app.

02a

Every feature, and what it does

Fourteen features across five roles: predict, decide, supply, detect, explain.

1. Demand forecasting PREDICT

What it does. Predicts how many litres the island will use, every hour, 48 hours ahead.

How it works. A RandomForestRegressor trained on two years of hourly history using five features: hour of day, day of week, weekend flag, temperature and tourist season. Evaluated on a chronological split so it is tested on the future.

Why it matters. Everything else depends on this. Without a demand curve you cannot know whether the tank will hold, so you cannot decide anything except reactively.

`ml/train_model.py, logic/demand.js` — MAE 45.3 L/h · R^2 0.987 · 89% better than predicting the average

2. Pump scheduling DECIDE

What it does. Decides each hour whether to run the desalination pump, and on what power.

How it works. Eight rules in strict priority order with a 12-hour lookahead that projects tank level against forecast demand, full-solar hours and expected rainfall. First matching rule wins.

Why it matters. This is the product. A wall timer runs the pump when power is most expensive; this runs it when power is free, and spends a little diesel early to avoid spending a lot later.

`logic/scheduler.js` — 78% less diesel than a fixed timer, zero shortage hours

3. Rainwater harvesting SUPPLY

What it does. Captures rainfall into the tank and holds the pump when a shower is coming.

How it works. Precipitation comes from the same Open-Meteo call as solar. harvest = rain_mm × catchment_m^2 × runoff. Rule 2 suppresses pumping when forecast rain exceeds forecast demand over the next six hours.

Why it matters. Funafuti is in reality almost entirely rainwater-fed, with desalination as the backup. A scheduler that ignores rain is optimising the wrong variable, and makes water it is about to spill.

`logic/scheduler.js — harvestFrom(), Rule 2` — 11.3 m^3 harvested — 24% of all water supplied

4. Battery storage SUPPLY

What it does. Stores surplus midday solar and spends it running the pump after dark.

How it works. Solar serves the pump first; surplus charges the battery at round-trip efficiency. When pumping without enough sun, the battery discharges before any diesel is burned. Energy that arrives with the battery full is counted as curtailed.

Why it matters. The demand peak is 18:00–20:00 and the solar peak is midday. Without storage those never meet and the evening runs on diesel.

`logic/scheduler.js — settleHour(), Rule 4` — 66 kWh discharged to the pump · 304 kWh curtailed

5. Storm preparation DECIDE

What it does. Abandons fuel economy and fills the tank to 100% before a cyclone lands.

How it works. Scans the 24-hour wind outlook. Above the storm threshold, Rule 0 overrides every cost rule until the tank is full.

Why it matters. After landfall the array is down, the pump may be offline and the resupply barge will not sail. Whatever is in the tank is what the island has. Optimising for fuel at that moment is the wrong objective.

`logic/scheduler.js — Rule 0, components/StormBanner.js` — Simulate cyclone control, since the live forecast has none

6. Leak detection DETECT

What it does. Finds a burst pipe from the pattern of the loss, hours before the tank looks low.

How it works. Each hour, compare the drop the sensor reports against the drop the forecast predicts (demand − pumped − harvested). σ is calibrated on the first twelve hours as a known-good baseline. Alert on 3σ for three consecutive hours.

Why it matters. A level threshold cannot tell a leak from a busy evening — by the time the tank is visibly low, thousands of litres are gone. A forecast knows what normal looks like at 03:00 on a Tuesday.

`logic/leak.js` — 450 L/h burst caught in 2 hours · estimated 452 L/h at 99% confidence

7. Panel soiling detection DETECT

What it does. Notices the solar array is producing less than the sunlight falling on it warrants.

How it works. Compares metered array output against the irradiance-derived prediction across daylight hours only, and reports the shortfall and the kWh not collected.

Why it matters. Salt spray and dust cost an island real diesel silently, for months. Nobody climbs up to check panels that are still producing something.

`logic/assets.js — detectSoiling()` — 13% below predicted across 20 daylight hours — 90 kWh lost

8. Membrane fouling prediction DETECT

What it does. Predicts the date the reverse-osmosis membranes will need cleaning.

How it works. Specific energy — kWh per m^3 of fresh water — rises as membranes foul, because the pump must push harder for the same flow. Least-squares fit on 90 days of daily readings, projected forward to the clean-in-place threshold.

Why it matters. Maintenance on an island is scheduled around a barge, not around a fault. Six weeks of warning is the difference between a planned clean and an emergency one.

`logic/assets.js — projectFouling()` — +11.1% from design · threshold reached in 40 days

9. Diesel inventory DETECT

What it does. Warns when the fuel will run out before the next resupply barge.

How it works. Projects the schedule's burn rate against litres remaining and days until the barge, and runs the same projection for both baseline strategies.

Why it matters. Burning too much diesel and running out of diesel are different failures. The barge does not come early because you ran out.

`logic/fuel.js` — AquaGrid 39 days; fixed timer dry 3.4 days early, reactive 4.4 days early

10. Demand-side advisory EXPLAIN

What it does. Tells residents the hours when using water costs the island nothing.

How it works. Finds runs of consecutive hours where the pump is already running on sun or stored sun, and turns them into a public notice — drafted by the LLM, or by a deterministic template offline.

Why it matters. Everything else schedules supply. The cheapest litre of diesel is the one nobody needed because the laundry ran at noon.

`logic/advisory.js — freeSolarWindows()` — Free windows surfaced as tappable chips with litres available

11. Rationing outlook EXPLAIN

What it does. Measures supply per person per day against the WHO minimum, and drafts the notice if it falls short.

How it works. Divides available supply over the horizon by population and days, and compares it to 15 L per person per day — the WHO floor for drinking, cooking and basic hygiene.

Why it matters. When shortfall is unavoidable, the question stops being about fuel and becomes about whether people have enough to drink. That deserves a number, not a vibe.

`logic/advisory.js — rationingOutlook()` — 18.2 L per person per day — 1.2× the WHO minimum

12. Any-island operation PREDICT

What it does. Points the whole system at any location on Earth, with no retraining.

How it works. The trained forest is evaluated exhaustively over its entire input grid and exported as a 15,120-cell lookup table. Open-Meteo geocoding resolves a typed name; the forecast endpoint supplies that island's real weather; the plant is scaled to its population.

Why it matters. A demo tied to one island is a demo. This is the difference between a case study and a system.

`logic/demand.js, logic/plant.js, lib/geo.js` — Table reproduces the model to a mean 5.5 L/h — 12% of its own error

13. Operator briefing and Q&A EXPLAIN

What it does. Writes the shift briefing in plain language and answers questions about the decisions.

How it works. A compact state object — never the whole forecast — goes to an open-weight model on Groq, raced against a six-second timeout. Three preset questions let a judge interrogate the scheduler in one tap.

Why it matters. A decision nobody can interrogate is not one an operator will trust. Every hour already carries a machine-written reason; this turns the set of them into something a person reads.

`lib/ai.js, components/BriefingCard.js, AskPanel.js` — ~620 ms round trip · falls back to a deterministic template

14. Offline operation PLATFORM

What it does. Runs completely with no network — scheduling, detection, charts and briefing.

How it works. The model's predictions and the weather forecast are bundled as a static asset at build time. The only two network calls, Groq and Open-Meteo geocoding, are optional enhancements that degrade to a deterministic path.

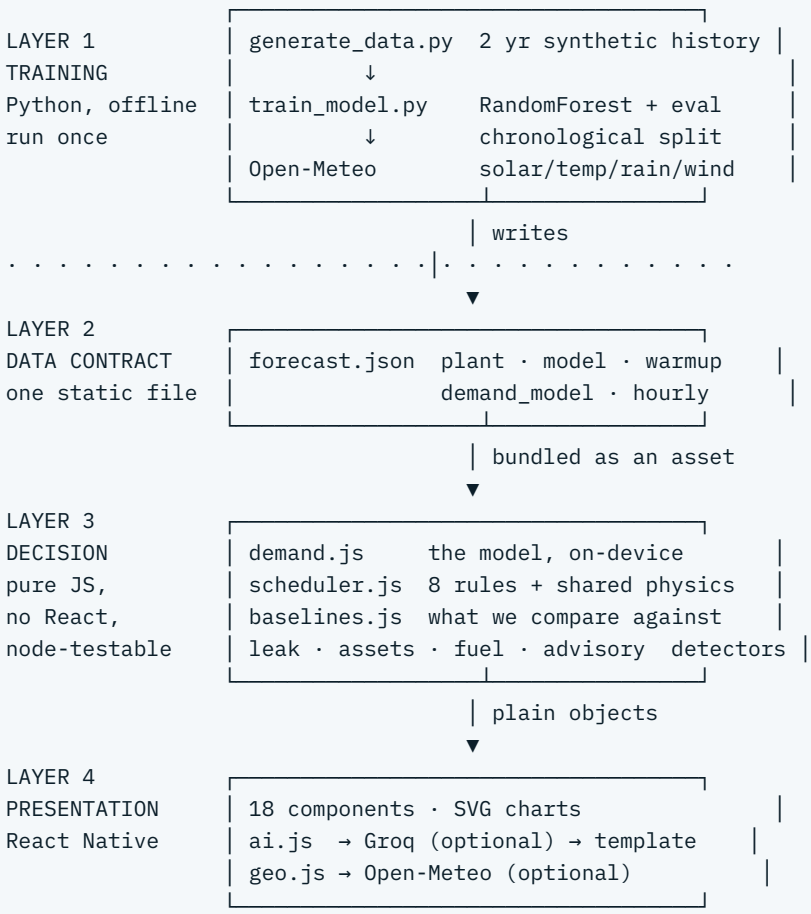
Why it matters. This is the actual operating condition on an isolated island. It is also why there is no backend to fail.

`assets/forecast.json, lib/ai.js fallback` — Airplane mode: every screen renders, only the briefing degrades

02b

The layered architecture

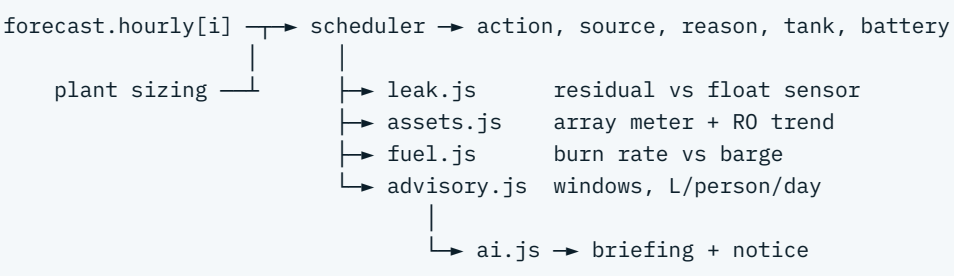
Four layers. Everything above the dotted line runs once on a laptop.



Why the layers are cut here

Layer 3 has **no React imports at all**, which is what makes `node src/logic/test.js` possible with no simulator and no phone. Layer 2 is a single file, so the Python and JavaScript halves can be developed and debugged independently — you can hand-edit `forecast.json` to reproduce a bug without touching Python. Layers 3 and 4 need no network; the two calls that do are strictly optional and degrade to a deterministic path.

Runtime data flow, one hour at a time



02c

The two AI systems

Different things. Only the first one makes decisions.

	DEMAND FORECASTER	NARRATOR
Model	`RandomForestRegressor`	`openai/gpt-oss-120b`
Type	Supervised regression, 180 trees, depth 14	Open-weight LLM, ~120B params
Trained by us?	**Yes** — 16,080 h, chronological split	No, used as-is
Runs where	Python offline, then as a lookup table on-device	Groq API, optional
Decides	**Everything.** Every pump hour, every alert	**Nothing.** Phrases what the scheduler decided
If it fails	The app cannot function	Template fallback, app unaffected
Verified by	MAE 45.3 L/h, R^2 0.987, 89% over naive	n/a

> SAY THIS IF A JUDGE ASKS WHAT THE AI IS
>
> The scheduler, the detectors and the savings are pure arithmetic over the forecaster's output. **Remove the LLM entirely and every number we quote is unchanged.** The intelligence is in the trained model and the rules, not in the prose. Judges routinely assume "AI" means the LLM; correcting that early is worth doing.

Technology stack

LAYER	TECHNOLOGY	VERSION	WHY THIS ONE
ML	scikit-learn	`1.9.0`	Interpretable, gives feature importances, no GPU
	pandas / numpy	`3.0.5` / `2.5.2`	Feature frames and synthetic history
	Python	`3.14.4`	
App	Expo SDK	`57.0.18`	Expo Go runs on a real phone, no native build or signing
	React Native	`0.86.3`	
	React	`19.2.3`	
	react-native-svg	`15.15.4`	**Only native module.** Charts hand-drawn
	Node	`25.6.1`	Runs the logic tests directly via ESM auto-detection
Weather	Open-Meteo Forecast	–	Radiation, temperature, precipitation, wind. No key
Geocoding	Open-Meteo Geocoding	–	Name → lat/lon/population. No key
LLM	Groq	–	Open-weight models, free, no card, ~620 ms
Hosting	GitHub Pages	–	The site. The app has no host
State	useState / useMemo	–	No Redux, no context. It is one screen

DELIBERATELY ABSENT

No backend, database, auth, navigation library, charting library, state-management library, CSS framework, or `eas build`. Each is a dependency that can break at hour four, and none earn their place.

Component inventory

MODULE	LINES	RESPONSIBILITY
`ml/train_model.py`	389	Train, evaluate, fetch weather, export lookup table + JSON
`ml/generate_data.py`	80	Two years of synthetic history
`logic/scheduler.js`	268	Eight rules, shared physics, warm-up replay
`logic/assets.js`	117	Panel soiling, RO membrane fouling
`logic/advisory.js`	98	Free-solar windows, WHO-minimum rationing
`logic/leak.js`	95	Burst-pipe residual detection, simulated sensor
`logic/baselines.js`	61	Fixed-timer and reactive strategies
`logic/fuel.js`	45	Diesel inventory against barge lead time
`logic/plant.js`	39	Population scaling
`logic/demand.js`	32	The exported model, on-device
`lib/ai.js`	189	Briefing, Q&A, notice — each with a fallback
`lib/geo.js`	155	Location search and live weather
`App.js`	239	Composition, demo switches, async effects
`components/` × 18	1,421	UI, SVG charts, all presentational

About **2,800 lines of JavaScript and 470 of Python.** The decision layer — everything the project actually claims — is 755 lines with no dependencies beyond the language.

03

Repository map

Every file that matters, one line each.

ml/ — the Python half

`generate_data.py`	Synthesises two years of hourly demand with real, readable structure. Auditable in one screen.
`train_model.py`	Trains the forest, evaluates on a chronological split, fetches weather, exports `forecast.json`.
`requirements.txt`	numpy, pandas, scikit-learn. Nothing else.

app/src/logic/ — pure JavaScript, no React, node-testable

`scheduler.js`	The eight rules, the shared physics in `settleHour`, and the warm-up replay.
`baselines.js`	Fixed-timer and reactive strategies. Same physics, different rule.
`leak.js`	Burst-pipe detection by residual analysis, plus the simulated tank sensor.
`assets.js`	Panel soiling and RO membrane fouling.
`fuel.js`	Diesel inventory against barge lead time.
`advisory.js`	Free-solar windows and litres per person per day.
`demand.js`	The exported model, running on-device.
`plant.js`	Scales a plant to an arbitrary population.
`test.js`	The definition-of-done harness. `node src/logic/test.js`.

app/src/lib/

`geo.js`	Open-Meteo geocoding and live weather for any island.
`ai.js`	Groq briefing, question answering and public notice, each with a deterministic fallback.

app/src/components/ — 16 components

Covered card by card in section 12.

Other

`app/assets/forecast.json`	The single interface between the two halves. Generated, not hand-edited.
`app/src/config.js`	**Gitignored.** Holds the Groq key. Copy from `config.example.js`.
`docs/index.html`	The project website. Charts generated from the real forecast.

04

The data contract — forecast.json

Everything the app renders comes from here. Import it directly; Metro bundles JSON with no config.

```
import forecast from './assets/forecast.json';
```

KEY	WHAT IT IS
`island`	Name, lat, lon, population.
`weather_source`	`"open-meteo (live)"` or the offline fallback label. Drives the header pill.
`plant`	Tank, pump, output, diesel rate, reserve floor, target, array, catchment, runoff, battery, fuel stock, barge interval, storm threshold.
`model`	MAE, RMSE, R^2, naive baseline, row counts, feature importances.
`demand_model`	The 15,120-cell lookup table. Section 10.
`maintenance`	90 days of RO specific-energy readings, plus design and clean threshold.
`validation_curve`	72 held-out hours, actual vs predicted, for the model detail modal.
`warmup`	The 24 hours *before* the horizon. Replayed to derive the starting tank level.
`hourly`	48 entries: time, hour, solar_kw, temp_c, cloud_pct, rain_mm, wind_kmh, predicted_demand_lph.

WHY WARMUP EXISTS

Without it every island opens at exactly the seeded percentage — 59% — and the biggest number on screen looks like a constant rather than a measurement. Replaying yesterday's real weather makes hour zero a consequence. Funafuti opens at 32%, Reykjavík at 41%.

05

The ML pipeline

`ml/generate_data.py and ml/train_model.py`

generate_data.py

Two years of hourly demand with structure a model can genuinely learn, not noise it can only memorise:

- Twin daily peaks — morning wash 06:00–08:00, evening cooking 18:00–20:00
- Weekends +30% — people are home, laundry day
- Tourist season (Jun–Aug) +22%
- Demand rises ~3.5% per °C
- Gaussian noise so nothing is perfectly predictable

Anyone can read this file and verify the patterns are real, which is exactly what you want a judge to be able to do.

train_model.py

`RandomForestRegressor(n_estimators=180, max_depth=14, min_samples_leaf=3)` on a **chronological split** — the last 60 days are held out, so the model is tested on the future, never on shuffled rows. This is the single most defensible thing about the model and you should say "chronological split" out loud.

METRIC	VALUE
MAE	**45.3 L/h**
RMSE	57.0 L/h
R^2	**0.987**
Naive "predict the mean"	409.7 L/h
Improvement over naive	**89%**

Feature importances: `hour` 0.897, `is_tourist_season` 0.032, `is_weekend` 0.031, `day_of_week` 0.028, `temp_c` 0.012. Hour dominating is correct — that is what a village water profile looks like.

The script then fetches Open-Meteo with `past_days=1&forecast_days=3&timezone=auto`, converts shortwave radiation to kW through the panel area, predicts demand forward, and writes the JSON.

> DO NOT CHANGE TIMEZONE=AUTO
>
> Funafuti is UTC+12. With UTC timestamps, peak solar lands at local midnight and *every* pump decision inverts. This was a real bug we already fixed.

06

The scheduler

`app/src/logic/scheduler.js — the core of the system`

Shared physics: settleHour()

Every strategy — ours and both baselines — applies its decision through the same function, so comparisons differ only in the rule. Order of operations per hour:

1. **Solar first.** `fromSolar = min(solar_kw, pumpKw)`
2. **Battery next.** Discharge before burning anything, capped at the pump draw.
3. **Diesel last.** `dieselL = remaining_kW × diesel_l_per_kwh`
4. **Charge.** Surplus solar charges the battery at round-trip efficiency; what will not fit is *curtailed* and counted.
5. **Water.** `tank + pumped + harvested − demand`, clamped to capacity. Overflow is counted as spill.
6. **Fuel stock** decrements. Shortage hours increment if the tank hits zero.

`harvest = rain_mm × roof_catchment_m2 × runoff_coeff`. One millimetre over one square metre is one litre.

The eight rules — strict priority, first match wins

#	RULE	FIRES WHEN
0	**Storm prep**	Cyclone-force wind within 24 h. Fill to 100% regardless of cost.
1	**Emergency**	Below the 25% reserve floor.
2	**Rain hold**	More rain than demand in the next 6 h, reserves comfortable. Hold; the tank fills free.
3	**Free solar**	Sun alone clears the full pump draw and the tank is below target.
4	**Stored solar**	Shortfall ahead and the battery can carry the pump.
5	**Pre-emptive hybrid**	Shortfall ahead with partial sun. Spend a little diesel now to avoid a lot tonight.
6	**Reluctant diesel**	Shortfall ahead, no sun, empty battery, reserves thin.
7	**Hold**	Everything else. Wait for the sun.

The lookahead

```
horizon        = next 12 hours
projDemand     = sum of predicted demand
projSolarHours = hours where solar_kw >= pumpKw
projRainL      = sum of rain × catchment
willBreach     = tank + (projSolarHours × outLph) + projRainL - projDemand < floorL
```

Rules 2, 4 and 5 are the intelligence of the system. They are the only ones acting on information the plant does not yet have. If none of them fire in a demo run, the plant parameters have drifted — see section 14.

warmUpTank()

Runs the scheduler over the `warmup` array and returns the final tank level, which becomes `initial_tank_l` for the displayed horizon.

07

Baselines

app/src/logic/baselines.js — what makes the savings number honest

Both run over the **identical forecast with identical physics** — same rain, same battery, same tank, same `settleHour`. Only the decision rule differs. A judge will ask how the savings were computed; this is the answer.

`fixedTimerSchedule`

Pumps 01:00–05:00 every day regardless of sun, rain or tank level. This is how these plants are *actually* run today — a wall timer set to mainland off-peak grid hours. On a solar island it is exactly backwards, because those hours have no sun.

`reactiveSchedule`

Pumps whenever the tank drops below 80%. Responds to the tank but is blind to the forecast, so it pumps at night as readily as at noon and makes water it is about to spill.

STRATEGY	DIESEL	COST	CO_2	PUMP H
AquaGrid	**15.3 L**	**$25**	**41 kg**	6
Fixed timer	70.0 L	$112	188 kg	8
Reactive	79.2 L	$127	212 kg	11

78% less diesel than the fixed timer. Note it delivers comparable water in *fewer* pump hours — it chooses which hours.

08

The four detectors

One idea: compare what the plant does against what the forecast says it should.

Burst pipe — leak.js

```
expectedDrop = demand - pumped - harvested + spilled
actualDrop   = sensor[i-1] - sensor[i]
residual     = actualDrop - expectedDrop     // positive = unexplained loss
sigma        = std(residual over first 12 h) // known-good baseline
ALERT when residual/sigma > 3 for 3 consecutive hours
```

A 450 L/h burst injected at hour 14 is caught at **hour 16**, estimated at 452 L/h with 99% confidence.

> DO NOT DROP HARVESTED FROM EXPECTEDDROP
>
> Rainwater is free inflow the forecast knows about. Omit it and every shower reads as a negative leak. This broke once already.

Dirty panels — assets.js / detectSoiling

Compares metered array output against the irradiance-derived prediction, over daylight hours only — at night both are zero and the ratio is meaningless. Reports shortfall percentage and kWh not collected. Currently: **13% below predicted, 90 kWh lost.**

Fouling membranes — assets.js / projectFouling

Specific energy (kWh per m^3) is the standard health metric for a reverse-osmosis train and rises as membranes foul, because the pump pushes harder for the same flow. Ordinary least squares on 90 days of daily readings, projected to the clean-in-place threshold. Currently: **+11.1% from design, threshold in 40 days.**

Fuel exhaustion — fuel.js

Burning too much diesel and running out of diesel are different failures.

STRATEGY	BURN	LASTS	VS 12-DAY BARGE
AquaGrid	8 L/day	39 days	holds
Fixed timer	35 L/day	8.6 days	**dry 3.4 d early**
Reactive	40 L/day	7.6 days	**dry 4.4 d early**

This is arguably a stronger claim than the percentage. The barge does not come early because you ran out.

09

Advisory — talking to the village

app/src/logic/advisory.js

Everything else schedules supply. This is the one piece that addresses demand. The cheapest litre of diesel is the one nobody needed.

`freeSolarWindows()`

Runs of consecutive hours where the pump is already running on sun or stored sun — hours when extra village demand costs the island nothing. Turned into chips in the UI and a sentence in the notice.

`rationingOutlook()`

Supply per person per day against the **WHO minimum of 15 L** for drinking, cooking and basic hygiene. Currently 18.2 L — 1.2× the floor, so no notice is warranted. When it falls below, the LLM drafts the public notice instead of the "when water is cheap" one.

The advisory has its own system prompt written for *residents*: no kilowatts, no plant jargon, say what to do and when.

10

Any location

app/src/logic/demand.js, plant.js and lib/geo.js

We ship the model, not only its predictions

Every feature the forecaster uses is categorical — hour, day of week, weekend, tourist season — except temperature, a single continuous variable. So the whole function can be evaluated exhaustively on a grid and shipped as a table.

15,120 cells: 2 seasons × 7 days × 24 hours × 45 temperature buckets at 0.5 °C. Shape is `table[tourist][dow][hour][tempBucket]`, in L/h for the reference population.

This is the model, not an approximation. The only loss is temperature discretisation, costing a **mean 5.5 L/h** — 12% of the model's own 45.3 L/h error. `test.js` asserts it stays below a quarter of the model MAE. `demand.js` interpolates between adjacent buckets because a forest is a step function in temperature.

plant.js — sizing for an arbitrary island

Tank, pump, output, panel area, roof catchment, battery and fuel storage all scale linearly with population off the reference plant. That holds the ratios that drive decisions constant — specific energy per m^3, days of storage, and how many hours a day solar clears the pump draw — so **only the weather genuinely differs between locations.** State this as an assumption; it is one.

geo.js

Open-Meteo geocoding, then the same forecast endpoint with `past_days=1&timezone=auto`. Both free, no key.

> SEARCH QUIRK, ALREADY HANDLED
>
> The geocoder matches settlement names only, so `"Malé Maldives"` and `"Apia, Samoa"` return nothing on a literal query. We retry on the first token and prefer results whose country or region matches the rest.

ISLAND	POPULATION	SAVEDWHY
Apia, Samoa	40,407	80%abundant sun
Malé, Maldives	103,693	70%equatorial, steady
Santorini, Greece	15,453	64%strong but seasonal
Reykjavík, Iceland	118,918	23%little sun to schedule

Reykjavík at 23% is the honest answer, not a bug. Where there is little sun there is little for a scheduler to win.

11

The LLM layer

`app/src/lib/ai.js`

Three functions, all with the same two-path shape:

`getBriefing`	The operator's shift briefing.
`askOperator`	Answers the three preset questions.
`getAdvisory`	The public water notice, in a different voice.

The template is the primary path — deterministic, instant, works in airplane mode. The Groq call is an enhancement raced against a 6-second timeout. Any error, timeout or missing key falls silently back, and the UI labels which one wrote the text so we never claim an LLM wrote a template.

Provider

Groq free tier, no credit card. Model is `openai/gpt-oss-120b` — **open-weight**, so no proprietary model is in the loop. `groq/compound-mini` is tried as a second model before the template.

> GPT-OSS IS A REASONING MODEL
>
> Without `reasoning_effort: 'low'` it spends the entire token budget thinking and returns *empty content*. We hit exactly this. Also note `llama-3.3-70b-versatile` is decommissioned on Groq — do not put it back.

> KEY HANDLING
>
> `src/config.js` is gitignored and no key is committed. The key lives client-side because there is no backend to hide it behind — the deliberate trade for an offline app. Say so in the README rather than hoping nobody notices. Beyond a demo this belongs behind a proxy, and the demo key should be revoked afterwards.

12

The screen, card by card

One SafeAreaView, one ScrollView, no navigation library.

COMPONENT	WHAT IT SHOWS
`Header`	Island, population, live/offline pill from `weather_source`, timestamp.
`LocationPicker`	Current island and a search sheet. **Anchored to the top** so the keyboard cannot cover the results.
`StormBanner`	Only when a cyclone is in the outlook. Explains why fuel rules are overridden.
`TankGauge`	Tank %, litres, hours of supply, tick marks at the 25% floor and 92% target.
`DecisionCard`	**The most important element.** Current action colour-coded by source, with the scheduler's own reason string, demand, production, rain inflow, diesel and battery draw.
`BriefingCard`	The operator briefing, badged `AI` or `on-device`.
`AskPanel`	Three preset questions. Presets not free text, so nobody types on camera.
`SolarDemandChart`	Solar area, demand line, rainfall bars, dashed rule at the pump draw. That rule makes the whole strategy legible.
`PumpTimeline`	24 bars coloured by source. Tap one to read that hour's reason.
`BatteryCard`	State of charge over 24 h, plus solar-to-pump, from-battery and curtailed kWh.
`WaterSourcesCard`	Rain vs desalinated split. On Tuvalu this is the headline.
`SavingsCard`	Three strategies, diesel, cost, CO_2. Hero percentage computed, never hardcoded.
`FuelCard`	Litres remaining, days to empty, barge countdown, and the same for both baselines.
`LeakPanel`	Green all-clear or red alert, plus **Simulate burst pipe**.
`AssetHealthCard`	Soiling and fouling, with the 90-day regression chart and **Simulate dirty panels**.
`AdvisoryCard`	Public notice, free-solar window chips, litres per person against the WHO marker.
`ModelCard`	MAE, R^2, 89% vs naive, feature-importance bars, and a modal with predicted vs actual.
`DemoBar`	**Simulate cyclone.** Every switch changes a real input; nothing fakes an outcome.

Charts are hand-drawn with `react-native-svg`. **Do not add a charting library** — version drift against the Expo SDK is a real risk and each chart is about sixty lines.

13

Testing and definition of done

One command, exits non-zero on failure.

```
cd app && node src/logic/test.js
```

It asserts: zero shortage hours, a clean run produces no leak alert, an injected burst *is* caught within three hours, savings are positive, and the lookup table stays within a quarter of the model's MAE. It prints the full comparison table, the 24-hour timeline and the warm-up result.

Before recording the video

- `python train_model.py` runs clean and prints MAE / R^2 / importances
- `forecast.json` has 48 hourly entries and 24 warm-up entries
- `node src/logic/test.js` passes
- The app loads in Expo Go on a physical phone
- Savings are computed from both schedules, not hardcoded
- "Simulate burst pipe" alerts within 3 simulated hours
- Airplane mode: every screen renders; only the briefing degrades, and it degrades to the template rather than an error
- No API key committed
- README quotes the metrics `train_model.py` actually printed

> EXPO GO DEV OVERLAY
>
> The floating blue gear button is Expo Go's, not ours. Drag it into a corner before recording.

Things you must not “fix”

Values tuned so the system has a real decision to make. Every one of these looks wrong and is not.

VALUE	WHY IT IS WHAT IT IS
`timezone=auto`	Funafuti is UTC+12. UTC timestamps put peak solar at local midnight and invert every decision.
`array_m2 = 320`	Peak output only clears the 50 kW pump draw for two or three midday hours. Enlarge it and every hour becomes a solar hour, the reserve floor never binds, and rules 4–6 become dead code.
`desal_pump_kw = 50`	Raised from 42 for the same reason. At 42 the scheduler coasted on solar alone and reported an implausible 100% saving.
`tank_capacity_l = 60000`	About 2.7 days of demand. Deliberately tight, so the reserve floor actually binds.
`roof_catchment_m2 = 900`	Rain covers ~26% of demand. At 2,500 it covered 72% and the scheduler had nothing to do.
`diesel_stock_l = 300`	25% of the day tank with 12 days to the barge. This is the state where a good scheduler stops being about money.
Fouling drift rate	Tuned so the clean-in-place date lands ~6 weeks out — near enough to act on, far enough to be a prediction rather than an alarm.

> THE GENERAL TRAP
>
> Three separate times, a change made the plant *too capable* and the scheduler stopped having to decide anything — which reads to a judge as a broken demo, not a good one. If you change a plant parameter, re-run `test.js` and confirm **all five sources still fire**: solar, battery, hybrid, diesel and hold.

Also do not

- Add a backend, navigation library, or charting library.
- Put a promise in `useMemo`.
- Hardcode the savings number. A judge will ask how it was computed.
- Reinstate `llama-3.3-70b-versatile`, which no longer exists on Groq.

Demo script

90 seconds, shot on the phone, one take, phone in a stand.

TIME	BEAT
0:00–0:10	**The problem.** “On an isolated island, desalination is the biggest power draw, and it usually runs on a dumb timer at night, on 100% diesel.”
0:10–0:22	**Open the app.** Tank gauge, current decision, read the reason line out loud. It is a full sentence, not a status code.
0:22–0:38	**Chart and timeline.** Point at the dashed pump-draw line. Green hours are free solar, cyan is stored solar running the pump after dark, amber is a deliberate partial-solar top-up.
0:38–0:48	**Rain.** “Tuvalu runs on rainwater; desal is the backup. A quarter of this water fell out of the sky, and the scheduler holds the pump when a shower is coming.”
0:48–0:58	**Savings, then fuel.** Say 78% out loud, then scroll to the fuel card: “and both naive schedulers run out of diesel before the next barge.”
0:58–1:10	**Tap Simulate burst pipe.** The alert fires. “It caught that from the forecast residual, not a threshold — no human would have noticed for hours.”
1:10–1:20	**Change location.** Search another island. Everything recomputes on-device.
1:20–1:30	**Airplane mode.** Everything still works. “Built for an island with no reliable connectivity.”

> ENDING
>
> Ending on airplane mode is the strongest closing beat available. Do not cut it.